

EE5203 L06 Study Sheet

GPIO Interrupts - Basri Erdogan

Essential question: How does one B1 press travel from a voltage change on PC13 all the way into your C code — without the main loop ever asking about the button?

What you should be able to do

- Walk the five-stage path with real names: PRESS → NOTICE (EXTI) → ASK (IRQ/NVIC) → LOOK UP (vector table) → RUN YOUR CODE (handler → callback).
- Configure a falling-edge EXTI pin in CubeMX and read the generated code.
- Explain IMR, FTSR, PR, and write-1-to-clear.
- Write a short callback that records an event in a *volatile* flag.
- Consume the event in `main()` and debounce with `HAL_GetTick()`.

The path, with real names

```
B1 press  -> PC13 falls      (active-low: released=1, pressed=0)
          -> EXTI13 edge detect (SYSCFG->EXTICR[3] field = 0010 routes PC13)
          -> pending bit set   (EXTI->PR bit 13 = 1)
          -> EXTI15_10_IRQn    (one request shared by lines 10-15)
          -> NVIC              (enable + priority, delivers to CPU)
          -> vector table      (IRQ number -> handler address)
          -> EXTI15_10_IRQHandler() (processor enters this)
          -> HAL_GPIO_EXTI_IRQHandler() (clears PR, then...)
          -> HAL_GPIO_EXTI_Callback(pin) (your code - HAL calls this)
```

Only one port can drive a given EXTI line at a time — the SYSCFG routing field stores exactly one choice.

Vocabulary table

Name	Job
EXTI	hardware edge detector; remembers the event in the pending bit
IRQ (EXTI15_10_IRQn)	the numbered request “this needs attention”
NVIC	enables, prioritizes, and delivers requests to the CPU
Vector table	maps each interrupt number to its handler’s address
Handler (...__IRQHandler)	the function the processor enters
Callback (..._Callback)	the function HAL calls; you write it

Configuration (CubeMX → generated code)

Three choices: PC13 = GPIO_EXTI13; falling edge + **no pull** (external R30); NVIC checkbox EXTI line[15:10] interrupts.

```
GPIO_InitStruct.Mode = GPIO_MODE_IT_FALLING;
GPIO_InitStruct.Pull = GPIO_NOPULL;
HAL_GPIO_Init(GPIOC, &GPIO_InitStruct); /* writes SYSCFG, IMR, FTSR */

HAL_NVIC_SetPriority(EXTI15_10_IRQn, 0, 0);
HAL_NVIC_EnableIRQ(EXTI15_10_IRQn);
```

Register, bit 13 = 1	Meaning	Who writes it
EXTI->IMR	line may raise an interrupt	HAL_GPIO_Init
EXTI->FTSR	falling edge selected	HAL_GPIO_Init
EXTI->PR	edge pending, unserviced	hardware sets; software clears by writing 1

Breakpoint inside the handler → PR bit 13 still reads 1 (the HAL helper has not cleared it yet).

The event-flag pattern

```
volatile uint8_t button_event = 0;           /* ISR-shared -> volatile */
uint32_t last_press_ms = 0;

void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    uint32_t now = HAL_GetTick();
    if (GPIO_Pin == GPIO_PIN_13 && (now - last_press_ms) >= 80) {
        last_press_ms = now;
        button_event = 1;                    /* record - nothing else */
    }
}

/* in while (1): */
if (button_event == 1) {
    button_event = 0;                        /* consume BEFORE acting */
    HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5); /* application work here */
}
```

- **Callback rules:** check the source, record, return fast. No `HAL_Delay`, no loops, no application logic.
- **volatile:** the interrupt changes the variable between two main-loop instructions; without it the compiler may keep using a stale copy.
- **Debounce:** contacts bounce for ~10 ms → several falling edges. 80 ms ignores all of them yet allows fast intentional presses. The unsigned subtraction `now - last_press_ms` stays correct across tick wrap-around.
- **Two different “flags”:** `EXTI->PR` bit 13 lives in hardware, cleared by the HAL helper (write-1-to-clear); `button_event` lives in your C program, cleared by `main()`.

SFR evidence

Claim	Evidence
PC13 routed to EXTI13	<code>SYSCFG->EXTICR[3]</code> line-13 field = 0010
falling edge selected	<code>EXTI->FTSR</code> bit 13 = 1
line unmasked	<code>EXTI->IMR</code> bit 13 = 1
edge awaiting service	<code>EXTI->PR</code> bit 13 = 1 while paused in the handler

Common mistakes

- Rising edge selected; internal pull fighting the board circuit.
- NVIC checkbox forgotten — EXTI sets `PR`, nothing is delivered.
- Callback compares the wrong pin mask.
- Long callback (delays/logic) blocking everything else.
- Missing `volatile` on the shared flag.
- No debounce — one press, several events.

Mastery self-check

- ☐ I can recite the path from press to callback with real names.
- ☐ I can state the three CubeMX choices and the registers they write.
- ☐ I can explain write-1-to-clear and where the pending bit lives.
- ☐ I can say which function the processor enters and which HAL calls.
- ☐ I can justify `volatile` in one sentence about stale copies.
- ☐ I can trace a bouncy press through the 80 ms guard.
- ☐ I can predict `EXTI->PR` bit 13 at a handler breakpoint.

Five review questions

1. The NVIC checkbox is left unchecked. Which registers still change on a press, and what never happens?
2. Why does the *release* of B1 not toggle the LED in the lab program?
3. A student calls `HAL_Delay(200)` inside the callback “for debouncing.” Give two distinct reasons this is wrong.
4. Edges arrive at $t = 5000, 5003, 5008, 5012, 5240, 5310$ ms. With the 80 ms guard, which are accepted, and what is `last_press_ms` at the end?

5. `button_event` works in a debug build but the LED stops responding at -O2, and adding `volatile` fixes it. Explain what the optimizer did.

See the L06 worksheet for extended practice. Worked solutions are released after the practice window.